Stroma World Spec — §3: Actors

> **Scope:** Every agent type that exists in the v1.0 simulation. If it's not listed here, it's not in the world.
> **Convention:** An "actor" is a cell type with autonomous behavior. Microenvironment fields (O_2, TGF-β, SHH, Drug, ECM) are not actors — they are the stage. **v1.0 count: 3 actors.** One tumor type, one stromal type with two modes, one inert boundary type.

Actor 1: PDAC Tumor Cell

Identity

The protagonist and the antagonist. This cell builds the barrier that protects it, creates the hypoxia that hardens it, and secretes the signals that recruit its own shield. Everything in the simulation exists because of what this cell does.

Goal (what the cell "wants")

Proliferate. Survive. Secrete. In that priority order.

The tumor cell has no explicit goal function — it doesn't optimize anything. Its behaviors are hardwired consequences of its genotype (KRAS=GOF, TP53=LOF, CDKN2A=LOF, SMAD4=LOF, BCL-XL=GOF). The "goal" framing is a modeling convenience: the cell acts as if it wants to grow and not die, because its broken checkpoints and survival machinery produce that outcome.

Core Behaviors

Behavior	Module	Description
Constitutive paracrine secretion	M2	Secretes TGF-β and SHH at rates set by Knobs 1a, 1b. Always on. Amplified under hypoxia (HIF1A=ON). This is the single behavior most responsible for the stroma barrier.

Behavior	Module	Description
Rapid proliferation	M4	Divides at a rate set by Knob 3a with checkpoints broken (Knob 3b ≈ 0). Paused only by severe mechanical compression (Module 8 → contact inhibition) or acute hypoxia.
Apoptosis resistance	M4	Low basal death rate (Knob 4 = 0.85). Dies only when drug reaches sufficient intracellular concentration AND efflux hasn't yet neutralized it.
Inducible EMT	M5	Transitions to mesenchymal phenotype when local TGF-β or hypoxia exceeds threshold (Knob 5a). Reversible. Changes motility, adhesion, MMP2 activity, and PD-L1 status.
Hypoxia adaptation	M1	Senses local O_2. When below threshold (Knob 6a), activates HIF1A → amplifies secretion, primes stress response, shifts toward resistant phenotype.
Drug-inducible efflux	M7	ABCB1 is silent at baseline. Activates via NRF2 after drug exposure + induction delay (Knob 7a). Reduces intracellular drug concentration by strength set by Knob 7b.
O_2 and drug consumption	Passive	Consumes O_2 (creates local depletion). Uptakes drug when exposed (acts as sink).

State Variables

Variable	Type	Initial value	Changes during run?	What reads it
gene_state[21]	Array of {WT, GOF, LOF}	Canonical PDAC genotype (see Gene State doc)	**No.** Fixed at cell birth.	Boolean network, all modules

Variable	Type	Initial value	Changes during run?	What reads it
hif1a_active	Boolean	OFF	Yes — toggles with local O_2	M2, M4, M5, M7
zeb1_active	Boolean	OFF	Yes — toggles with TGF-β + hypoxia signals	M4 (go-vs-grow), M6 (MMP2)
cdh1_expressed	Boolean	ON	Yes — inverse of ZEB1	M4 (adhesion)
nrf2_active	Boolean	OFF	Yes — activates under drug/oxidative stress	M7 (ABCB1 gating)
abcb1_active	Boolean	OFF	Yes — activates after NRF2 + delay	M4 (death resistance), M7 (efflux)
intracellular_drug	Continuous (0–1)	0	Yes — accumulates from uptake, decays from efflux	M7 (kill signal → M4)
time_since_drug_exposure	Continuous (hours)	NULL (no exposure)	Yes — starts counting at first drug contact	M7 (induction delay clock)
emt_extent	Categorical {NONE, LOW, MED, HIGH}	NONE	Yes — set by M5 based on signal strength + Knob 5b	M4, M5, M6

Variable	Type	Initial value	Changes during run?	What reads it
mechanical_pressure	Continuous (relative)	0	Yes — updated by PhysiCell mechanics + M8	M4 (contact inhibition)
position	(x, y) or (x, y, z)	Seeded per initial conditions	Yes — cell moves (motility), pushed (mechanics)	All modules (spatial sampling of fields)

What the tumor cell does NOT track in v1.0:

- Metabolite concentrations (glucose, lactate) — approximated by O_2 as proxy
- Immune checkpoint ligand expression level — tracked qualitatively via ZEB1 state, no immune cells to interact with
- Mutation history — no clonal evolution; genotype is fixed
- Cell age or replicative senescence — not modeled in v1.0

Actor 2: Pancreatic Stellate Cell / Cancer-Associated Fibroblast

Identity

One cell type, two modes. The quiescent pancreatic stellate cell (PSC) is the resting state. The activated cancer-associated fibroblast (CAF) is the barrier-building state. The transition between them is the central event of stroma formation.

This is the construction worker. The tumor gives orders (TGF-β, SHH). This cell builds the wall.

Goal (what the cell "wants")

As PSC: Nothing. Sit quietly. Wait for activation signals. **As CAF:** Produce ECM. Proliferate. Secrete TGF-β (positive feedback). Contract surrounding matrix.

Core Behaviors

Behavior	Module	PSC mode	CAF mode
Signal sensing	M3	Samples local TGF-β and SHH. Compares to activation threshold.	Already activated. GLI1 tracks ongoing SHH exposure.
Activation transition	M3	When combined TGF-β + SHH signal exceeds threshold → irreversible switch to CAF mode.	N/A (already activated; no reversion in v1.0).
ECM production	M6	None.	Produces HA (via HAS2) and collagen (via COL1A1) at high rates. Boosted if GLI1=ON.
TGF-β feedback secretion	M2	None.	Secretes TGF-β at moderate rate, creating the positive feedback loop that amplifies barrier formation.
Proliferation	M4	Minimal (near-quiescent).	Moderate. Boosted if GLI1=ON. Slower than tumor cells.
Contractility	M8 (passive)	None.	Exerts mild traction on surrounding ECM, contributing to matrix compaction and alignment. Modeled as a contribution to local ECM density rather than as an explicit force in v1.0.

Behavior	Module	PSC mode	CAF mode
Drug indifference	M7	Not a drug target.	Not a drug target. Standard chemotherapy in PDAC targets tumor cells, not CAFs. CAFs persist through treatment, which is why the barrier rebuilds after drug withdrawal.

State Variables

Variable	Type	Initial value (PSC)	Changes during run?	What reads it
acta2_active	Boolean	OFF	Yes — **once.** OFF → ON is irreversible in v1.0.	M2, M4, M6 (gates all CAF behaviors)
gli1_active	Boolean	OFF	Yes — tracks SHH exposure. Reversible (drops if SHH disappears).	M4 (proliferation boost), M6 (ECM boost)
activation_mode	Enum {PSC, CAF}	PSC	Yes — transitions to CAF when ACTA2 flips ON.	All modules (behavioral routing)
ecm_production_rate	Categorical {NONE, HIGH, BOOSTED}	NONE	Yes — NONE (PSC), HIGH (CAF), BOOSTED (CAF + GLI1=ON)	M6
tgfb_secretion_active	Boolean	OFF	Yes — ON when CAF mode	M2

Variable	Type	Initial value (PSC)	Changes during run?	What reads it
position	(x, y) or (x, y, z)	Seeded per initial conditions	Yes — mild motility when activated; less motile than tumor cells	All modules

What PSC/CAFs do NOT track in v1.0:

- Subtypes (myCAF vs. iCAF vs. apCAF) — all activated stellate cells are treated as one CAF phenotype. Subtype heterogeneity is Cycle 2+.
- SMAD4 status — stromal cells are SMAD4-intact. TGF-β fully activates them. This is not a variable; it's a fixed assumption.
- Drug response — CAFs are not targeted by the drug substrate. They survive treatment passively, which is biologically accurate for standard PDAC chemotherapy and critical for barrier rebuilding (Anchor 7, Anchor 10).
- Apoptosis — CAF death rate is near-zero in v1.0. Activated CAFs persist indefinitely. This simplification is acceptable because CAF turnover is slow relative to simulation timescales and CAF depletion strategies are not in the v1.0 EA intervention space.

The Activation Transition in Detail

This is the most important state change any actor undergoes in the simulation:

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PSC (quiescent)                    CAF (activated)
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ACTA2 = OFF                        ACTA2 = ON
ECM production = 0          ──→    ECM production = HIGH
TGF-β secretion = 0       (one-way)  TGF-β secretion = ON
Proliferation = minimal            Proliferation = moderate
Contractility = 0                  Contractility = active
GLI1 = OFF                         GLI1 = tracks SHH

Trigger: local TGF-β + SHH > threshold
Reversibility: NONE in v1.0
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The irreversibility is a deliberate v1.0 simplification. In reality, some CAF plasticity exists (particularly iCAF ↔ myCAF switching), but modeling reversible activation introduces oscillatory dynamics that complicate debugging without changing the qualitative barrier-formation outcome. If Reality Anchors 1 and 7 pass with irreversible activation, there's no need to add reversion yet.

Actor 3: Boundary Voxel (Inert Source)

Identity

Not a cell. A fixed boundary condition. Represents the vascular supply and tissue edge that the simulation domain is carved from. Exists only at the domain perimeter.

Goal

Maintain constant substrate concentrations at the domain edge. This actor is the simulation's connection to the outside world — the reason O_2 and drug can enter, and the implicit representation of vasculature that we're not modeling explicitly in v1.0.

Core Behaviors

Behavior	Description
O_2 supply	Maintains O_2 concentration at a fixed Dirichlet boundary value. Represents the vascular O_2 tension at the tissue edge. The interior O_2 profile is determined by diffusion through ECM and consumption by cells.
Drug delivery	When drug is applied (by the EA or by a test scenario), drug concentration is set at the boundary and diffuses inward. The boundary is the "vasculature" — drug enters from outside and must penetrate the stroma to reach the tumor.
TGF-β / SHH sink	Boundary acts as a mild sink for paracrine signals, preventing unrealistic accumulation at the domain edge. Signals produced by tumor cells diffuse outward and are partially absorbed at the boundary (Dirichlet = 0 or low value).
ECM = 0 at boundary	No stroma at the domain edge. ECM density is zero at the boundary and increases toward the tumor center. This enforces the correct gradient direction.

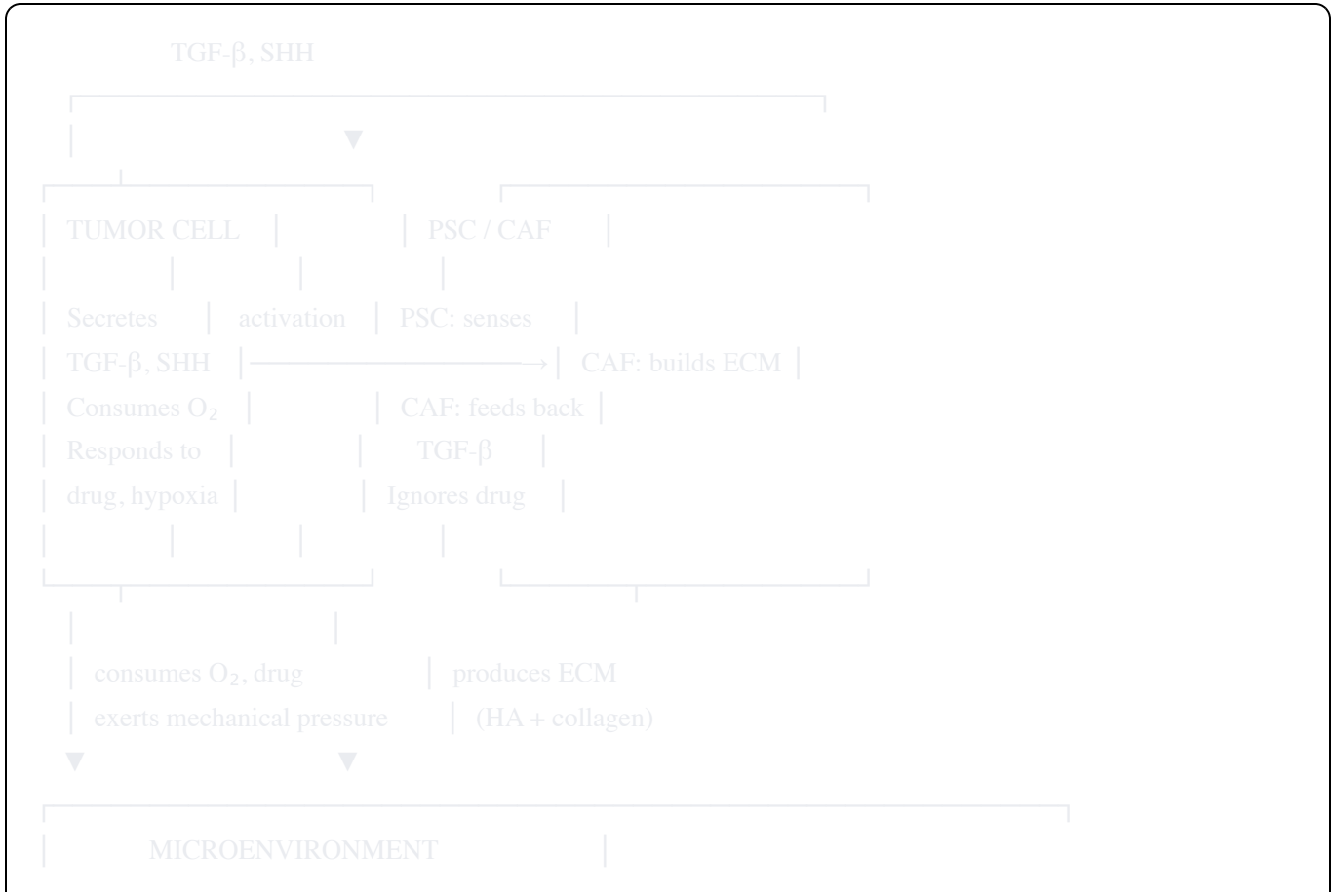
State Variables

Variable	Type	Value	Changes?
o2_boundary	Continuous	Fixed (literature value for peri-pancreatic tissue O_2 tension)	No, unless EA modifies delivery
drug_boundary	Continuous	0 at baseline; set by EA or scenario when drug is applied	Only when intervention begins
tgfb_boundary	Continuous	0 or near-0 (sink)	No
shh_boundary	Continuous	0 or near-0 (sink)	No

Variable	Type	Value	Changes?
ecm_boundary	Continuous	0 (no stroma at edge)	No

Why this actor exists: Without explicit vasculature (Cycle 3+), the boundary condition *is* the vasculature. Every substrate that enters the domain enters from the boundary. Every Reality Anchor that involves drug delivery (2, 7, 8, 9, 10) depends on the boundary actor correctly representing the drug source. The simplification is that drug delivery is spatially uniform around the perimeter — in reality, vessel density varies, but this heterogeneity is out of scope.

Actor Interaction Summary



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│ O₂ │ TGF-β │ SHH │ Drug │ ECM_density │
└──────────────────────────────────────────┘
                │
        supplied from
                │
        ┌──────────────────┐
        │ BOUNDARY VOXEL   │
        │ O₂ = fixed       │
        │ Drug = EA-controlled │
        │ TGF-β, SHH = sink │
        │ ECM = 0          │
        └──────────────────┘
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Who Talks to Whom

Source	Target	What passes	Edge refs
Tumor	Stroma	TGF-β, SHH (paracrine)	E06, E07
Stroma (CAF)	Environment	ECM (HA, collagen)	E10, E11
Stroma (CAF)	Tumor (indirect)	TGF-β feedback → more stroma → more hypoxia	E12, E20
Environment	Tumor	O_2 (sensed), Drug (uptaken), TGF-β (EMT signal)	E01, E08, E24
Environment	Stroma	TGF-β (activation signal), SHH (GLI1 signal)	E06, E07
Tumor	Environment	O_2 consumption, drug consumption, mechanical pressure	E14, passive
Boundary	Environment	O_2 supply, drug supply, signal absorption	Boundary conditions

Who Does NOT Talk to Whom

Non-interaction	Why
Tumor ↔ Tumor (direct signaling)	No juxtacrine or direct cell-cell signaling between tumor cells in v1.0. All communication is via microenvironment fields. Contact effects (adhesion, contact inhibition) are handled by PhysiCell mechanics, not by explicit signaling.
Stroma ↔ Stroma (direct signaling)	Same. CAFs don't signal each other directly. Coordination emerges from shared field exposure.
Drug → CAF	Standard PDAC chemotherapy does not target CAFs. CAFs are drug-indifferent. This is why the barrier rebuilds.
Immune cells → Anyone	No immune actors in v1.0. IET is tracked as bookkeeping but has no functional target.
Boundary → Cells (direct)	The boundary doesn't push, pull, or signal cells. It only sets field values. Cells that wander to the boundary are handled by boundary conditions (reflection or absorption, per domain geometry spec).

Actor Budget

Actor	Typical count in simulation	Computational cost driver
Tumor cells	500 – 5,000 (growing)	Proliferation creates new agents; most numerous by endgame

Actor	Typical count in simulation	Computational cost driver
PSC/CAFs	200 – 2,000 (seeded)	CAF activation is cheap; ECM production writes to field; moderate cost
Boundary voxels	Fixed (domain perimeter)	Zero per-step cost; set once at initialization

Total agent count at peak: ~3,000 – 7,000 cells. This is well within PhysiCell's comfortable range on a single LARCC node. Agent count is not the computational bottleneck — PDE solving for 5 microenvironment fields is.

What v1.0 Deliberately Excludes as Actors

Excluded actor	When it enters	Why not now
CD8+ T cells	Cycle 2 (immune evasion)	No immune system in stroma-only world. IET bookkeeping prepares for their arrival.
Macrophages (TAMs)	Cycle 2	Major immune-stroma crosstalk, but requires immune module.
Regulatory T cells	Cycle 2	TGF-β → Treg induction is tracked in IET but has no target cell.
Endothelial cells / vessels	Cycle 3 (vasculature)	O_2 and drug delivery approximated by boundary conditions.
Pericytes	Cycle 3	Vessel stability and anti-angiogenic therapy are out of scope.
Neurons / nerve cells	Not planned	Neural invasion is a PDAC feature but not a barrier-formation driver.

Excluded actor	When it enters	Why not now
Normal ductal epithelium	Not planned for v1.0	Tumor is pre-established; no modeling of initiation or PanIN progression. Could be added as a spatial boundary if needed.

Each excluded actor has a defined insertion point in the module architecture. The 8 modules and 28 edges were designed to accommodate Cycle 2 and 3 actors without rewiring — new actors plug into existing modules (e.g., CD8+ T cells would read IET state from tumor cells and ECM density from the environment, interacting with Modules 4, 5, and 6).

Three actors. One builds the prison (tumor). One is the prison (stroma). One is the outside world (boundary). Everyone else arrives in later cycles.